

Relay Catalysis for Selective Aerobic Oxidative Esterification of Primary Alcohols with Methanol

Yibo Yu,^a Jie Lin,^{†,a} Anni Qin,^{†,a} Huanan Wang,^{†,a} Jie Wang,^{†,a} Weiyi Wang,^{†,a} Guolin Wu,^{†,a} Qian Zhang,^{†,a} Hui Qian^{*,a} and Shengming Ma^{*,a,b}

^a Research Center for Molecular Recognition and Synthesis, Department of Chemistry, Fudan University, 220 Handan Lu, Shanghai 200433, P. R. China.

^b State Key Laboratory of Organometallic Chemistry, Shanghai Institute of Organic Chemistry, Chinese Academy of Sciences, 345 Lingling Lu, Shanghai 200032, P. R. China.

[†] These authors contributed equally.

Supporting Information

General Information	S2
Identification of the slowly oxidizing alcohols	S3
Optimization for the aerobic oxidative methyl esterification of alcohols	S4-S7
Experimental details and analytical data	S8-S31
References	S32-S33
¹ H, ¹³ C NMR and HPLC spectra of products	S34-S124

General Information. NMR spectra were taken with an Agilent-400 spectrometer (400 MHz for ^1H NMR, 100 MHz for ^{13}C NMR, 376 MHz for ^{19}F NMR). All reactions were carried out in 50 mL flask unless otherwise mentioned. $\text{Fe}(\text{NO}_3)_3 \cdot 9\text{H}_2\text{O}$ was purchased from Energy Chemical; TEMPO (98%) was purchased from Shanghai Darui Fine Chemical Co., Ltd.; BiCl_3 was purchased from Alfa Aesar; DCE and MeOH were purchased from Sinopharm Chemical Reagent Co., Ltd and Shanghai Titan Scientific Co., Ltd, respectively, and used directly as received without further treatment. All the temperatures were referred to the oil baths or ice-water bath (0 °C) used. The HRMS analysis was performed with Agilent 7200B GC/Q-TOF.

TEMPO: 2,2,6,6-Tetramethylpiperidinoxy

4-NHAc-TEMPO: 4-(Acetylamino)-2,2,6,6-tetramethylpiperidinoxy

4-OMe-TEMPO: 4-Methoxy -2,2,6,6-tetramethylpiperidinoxy

4-OH-TEMPO: 4-Hydroxy)-2,2,6,6-tetramethylpiperidinoxy

ABNO: 9-Azabicyclo[3.3.1]nonane *N*-oxyl

Identification of the slowly oxidizing alcohols

To a 50 mL flask were added $\text{Fe}(\text{NO}_3)_3 \cdot 9\text{H}_2\text{O}$ (0.06 mmol), TEMPO (0.05 mmol), BiCl_3 (0.1 mmol), **1a** (1.0 mmol), **2** (5.0 mmol), and DCE (3 mL) sequentially under air atmosphere. The reaction was then carried out at 50 °C under the atmosphere of O_2 from a balloon (about 2.5 L) for 48 h. The resulting mixture was cooled to 0 °C with an ice-water bath for 10 min, then warmed to rt. CH_2Br_2 (35 μL) was added as the internal standard. About 0.1 mL of the resulting mixture was taken and filtrated through a disposable syringe filter (PTFE, 13 mm, 0.45 μm) eluted with CDCl_3 (0.5 mL), and analyzed via ^1H NMR. Then the resulting mixture was collected and filtrated through a short column of silica gel eluted with diethyl ether (3 x 25 mL), concentrated, and analyzed via ^1H NMR with CH_2Br_2 (35 μL) as the internal standard. According to the comparison of two ^1H NMR spectra, the yields of products that have low boiling point were calculated.

Table S1. Identification of the slowly oxidizing alcohols^a

(A) Identification of the slowly oxidizing alcohols.

identify the slowly oxidizing alcohols

Entry	$\text{R}'\text{-CH}_2\text{OH}$	Recovery of 1a	NMR yield					
			3a	8	10	6	5a	11
1	BnOH (2a)	0	69%	86% (8a)	3% (10a)	0	10%	0
2	<i>n</i> -BuOH (2b)	11%	51%	32% (8b)	14% (10b)	19% ^b (6ab + 5a + 11ba)		
3	<i>n</i> -PrOH (2c)	9%	51%	14% (8c)	12% (10c)	8% (6ac)	12% ^c (5a + 11ca)	
4	EtOH (2d)	0	63%	12% (8d)	14% (10d)	14% (6ad)	9%	8% (11da)
5	MeOH (2e)	0	9%	ND ^d	9% (10e)	83% (6ae)	3%	1% (11ea)
6	<i>i</i> -PrOH (2f)	49%	24%	45% ^e (8f)	-	0	3%	-

^a The reaction was conducted with **1a** (1.0 mmol), $\text{R}'\text{CH}_2\text{OH}$ (5 mmol), $\text{Fe}(\text{NO}_3)_3 \cdot 9\text{H}_2\text{O}$ (6 mol%), TEMPO (5 mol%), and BiCl_3 (10 mol%) in 3 mL of DCE at 50 °C for 48 h with an O_2 balloon in a 50 mL flask. The NMR yield and recovery were determined by ^1H NMR analysis using dibromomethane as the internal standard. ^b Combined yield of **6ab**, **5a** and **11ba** based on alcohol **1a**. ^c Combined yield of **5a** and **11ca** based on alcohol **1a**. ^d ND. = not determined. ^e NMR yield of acetone **8f**.

Optimization for the oxidative methyl esterification of alcohols

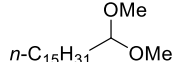
To a 25 mL or 50 mL flask were added Fe(NO₃)₃•9H₂O, TEMPO, Lewis acid, **1a**, MeOH, and DCE sequentially in open air. The reaction was then carried out at the specified temperature under the atmosphere of O₂ from a balloon (about 2.5 L) until completion of the reaction as monitored by TLC. The resulting mixture was filtrated through a short column of silica gel eluted with diethyl ether (3 x 25 mL), concentrated, and analyzed via ¹H NMR with CH₂Br₂ (35 μL) as the internal standard.

Table S2. Optimization of the amount of catalyst^a

$n\text{-C}_{15}\text{H}_{31}\text{CH}_2\text{OH}$ + MeOH 1a (1.0 mmol) 2e (5.0 equiv)		$\xrightarrow[\text{DCE, 50 } ^\circ\text{C, 40 h}]{\text{Fe(NO}_3)_3\cdot 9\text{H}_2\text{O (x mol\%)} \\ \text{TEMPO (y mol\%)} \\ \text{BiCl}_3 \text{ (z mol\%)}} \\ \text{Air + pure O}_2 \text{ balloon}$		$n\text{-C}_{15}\text{H}_{31}\text{CHO} + \text{HCOO}n\text{-C}_{16}\text{H}_{33} + n\text{-C}_{15}\text{H}_{31}\text{COOMe}$ 3a 11ea 6ae $n\text{-C}_{15}\text{H}_{31}\text{COO}n\text{-C}_{16}\text{H}_{33} + n\text{-C}_{15}\text{H}_{31}\text{C}(\text{OMe})_2$ 5a 7ae			
Entry	<i>x/y/z</i>	Recovery of 1a	NMR Y of 3a	NMR Y of 11ea	NMR Y of 6ae	NMR Y of 5a	NMR Y of 7ae
1	5/5/10	0	20%	1%	65%	4%	8%
2	6/5/10	0	17%	1%	74%	2%	6%
3	7/5/10	0	13%	1%	76%	4%	9%
4	8/5/10	0	14%	1%	77%	2%	4%
5	9/5/10	0	13%	1%	74%	2%	4%
6	10/5/10	0	18%	1%	66%	4%	6%
7	10/10/10	0	17%	2%	61%	6%	7%
8	8/3/10	12%	34%	1%	8%	4%	20%
9	8/7/10	0	14%	1%	77%	2%	2%
10	6/5/8	0	18%	1%	66%	4%	6%
11	6/5/6	0	22%	1%	64%	2%	8%

^a The reaction was conducted with **1a** (1.0 mmol), MeOH (5.0 mmol), Fe(NO₃)₃•9H₂O (x mol%), TEMPO (y mol%), and BiCl₃ (z mol%) in 4 mL of DCE at 50 °C for 40 h with an O₂ balloon in a 25 mL flask. The NMR yield and recovery were determined by ¹H NMR analysis using dibromomethane as the internal standard.

Table S3. Optimization of the amount of MeOH^a

$n\text{-C}_{15}\text{H}_{31}\text{CH}_2\text{OH}$ + MeOH 1a (1.0 mmol) 2e (<i>x</i> equiv)			$\xrightarrow[\text{DCE, 50 } ^\circ\text{C, 40 h}]{\text{BiCl}_3 \text{ (10 mol\%)} \\ \text{TEMPO (5 mol\%)} \\ \text{Fe(NO}_3)_3 \cdot 9\text{H}_2\text{O (6 mol\%)}}$		$n\text{-C}_{15}\text{H}_{31}\text{CHO}$ + $\text{HCOO}n\text{-C}_{16}\text{H}_{33}$ + $n\text{-C}_{15}\text{H}_{31}\text{COOMe}$ 3a 11ea 6ae		
			Air + pure O_2 balloon		$n\text{-C}_{15}\text{H}_{31}\text{COO}n\text{-C}_{16}\text{H}_{33}$ + 		
			DCE, 50 $^\circ\text{C}$, 40 h		5a 7ae		
Entry	<i>x</i>	Recovery of 1a	NMR Y of 3a	NMR Y of 11ea	NMR Y of 6ae	NMR Y of 5a	NMR Y of 7ae
1	3	0	26%	1%	65%	4%	3%
2	4	0	22%	1%	71%	4%	5%
3	5	0	14%	1%	74%	4%	8%
4	6	0	15%	1%	75%	4%	7%
5	7	0	15%	1%	74%	0	10%
6	8	0	16%	1%	67%	2%	14%

^a The reaction was conducted with **1a** (1.0 mmol), MeOH (x mmol), Fe(NO₃)₃•9H₂O (6 mol%), TEMPO (5 mol%), and BiCl₃ (10 mol%) in 4 mL of DCE at 50 °C for 40 h with an O₂ balloon in a 25 mL flask. The NMR yield and recovery were determined by ¹H NMR analysis using dibromomethane as the internal standard.

Table S4. Optimization of the concentration^a

$n\text{-C}_{15}\text{H}_{31}\text{CH}_2\text{OH}$ + MeOH				$\xrightarrow[\text{DCE (x mL), 50 } ^\circ\text{C, t h}]{\text{BiCl}_3 \text{ (10 mol\%)} \\ \text{TEMPO (5 mol\%)} \\ \text{Fe(NO}_3)_3 \cdot 9\text{H}_2\text{O (6 mol\%)}}}$		$n\text{-C}_{15}\text{H}_{31}\text{CHO}$ + $\text{HCOO}n\text{-C}_{16}\text{H}_{33}$ + $n\text{-C}_{15}\text{H}_{31}\text{COOMe}$					
1a (1.0 mmol)		2e (5.0 equiv)		Air + pure O_2 balloon		3a		11ea		6ae	
						$n\text{-C}_{15}\text{H}_{31}\text{COO}n\text{-C}_{16}\text{H}_{33}$		+ $n\text{-C}_{15}\text{H}_{31}\text{C}(\text{OMe})_2$			
						5a		7ae			
Entry	x	t	Recovery of 1a	NMR Y of 3a	NMR Y of 11ea	NMR Y of 6ae	NMR Y of 5a	NMR Y of 7ae			
1	2	40 h	0	27%	0	36%	4%	16%			
2	3	40 h	0	15%	1%	75%	4%	6%			
3	3	48 h	0	8%	1%	84%	4%	5%			
4	3.5	40 h	0	15%	1%	76%	4%	6%			
5	4	40 h	0	14%	1%	74%	4%	8%			
6	5	40 h	0	22%	1%	63%	4%	12%			
7	6	40 h	0	28%	1%	53%	4%	17%			

^a The reaction was conducted with **1a** (1.0 mmol), MeOH (5.0 mmol), Fe(NO₃)₃•9H₂O (6 mol%), TEMPO (5 mol%), and BiCl₃ (10 mol%) in x mL of DCE at 50 °C for t h or 48 h with an O₂ balloon in a 25 mL flask. The NMR yield and recovery were determined by ¹H NMR analysis using dibromomethane as the internal standard.

Table S5. Optimization of the reaction temperature^a

$n\text{-C}_{15}\text{H}_{31}\text{CH}_2\text{OH}$ + MeOH		$\xrightarrow[\text{Air + pure O}_2 \text{ balloon}]{\text{BiCl}_3 (10 \text{ mol\%})$ $\text{Fe(NO}_3)_3 \cdot 9\text{H}_2\text{O} (6 \text{ mol\%})$ $\text{DCE, } T^\circ\text{C, 48 h}$		$n\text{-C}_{15}\text{H}_{31}\text{CHO} + \text{HCOO}n\text{-C}_{16}\text{H}_{33} + n\text{-C}_{15}\text{H}_{31}\text{COOMe}$ 3a 11ea 6ae $n\text{-C}_{15}\text{H}_{31}\text{COO}n\text{-C}_{16}\text{H}_{33} + $ 5a 7ae			
1a (1.0 mmol)	2e (5.0 equiv)						
Entry	T	Recovery of 1a	NMR Y of 3a	NMR Y of 11ea	NMR Y of 6ae	NMR Y of 5a	NMR Y of 7ae
1	25	0	8%	1%	52%	4%	25%
2	35	0	7%	1%	76%	4%	9%
3	40	0	10%	1%	80%	4%	5%
4	45	0	11%	1%	81%	4%	7%
5	50	0	8%	1%	84%	4%	5%
6	55	0	23%	1%	60%	4%	7%
7	60	0	31%	1%	56%	4%	5%

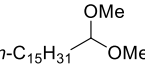
^a The reaction was conducted with **1a** (1.0 mmol), MeOH (5.0 mmol), Fe(NO₃)₃•9H₂O (6 mol%), TEMPO (5 mol%), and BiCl₃ (10 mol%) in 3 mL of DCE at T °C for 48 h with an O₂ balloon in a 25 mL flask. The NMR yield and recovery were determined by ¹H NMR analysis using dibromomethane as the internal standard.

Table S6. Optimization of the solvent^a

$n\text{-C}_{15}\text{H}_{31}\text{CH}_2\text{OH}$ + MeOH		$\xrightarrow[\text{Air + pure O}_2 \text{ balloon}]{\text{BiCl}_3 (10 \text{ mol\%})$ $\text{Fe(NO}_3)_3 \cdot 9\text{H}_2\text{O} (6 \text{ mol\%})$ solvent, 50 °C, 48 h}		$n\text{-C}_{15}\text{H}_{31}\text{CHO} + \text{HCOO}n\text{-C}_{16}\text{H}_{33} + n\text{-C}_{15}\text{H}_{31}\text{COOMe}$ 3a 11ea 6ae $n\text{-C}_{15}\text{H}_{31}\text{COO}n\text{-C}_{16}\text{H}_{33} + $ 5a 7ae			
1a (1.0 mmol)	2e (5.0 equiv)						
Entry	solvent	Recovery of 1a	NMR Y of 3a	NMR Y of 11ea	NMR Y of 6ae	NMR Y of 5a	NMR Y of 7ae
1	DCE	0	8%	1%	84%	4%	5%
2 ^b	DCM	4%	38%	2%	9%	4%	40%
3	CHCl ₃	0	43%	1%	25%	0	31%
4	CH ₃ CN	10%	27%	1%	9%	2%	23%
5	THF	48%	22%	0	0	0	0
6	Toluene	0	30%	2%	44%	0	15%
7	1,4-dioxane	0	20%	2%	21%	0	1%

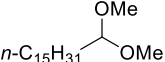
^a The reaction was conducted with **1a** (1.0 mmol), MeOH (5.0 mmol), Fe(NO₃)₃•9H₂O (6 mol%), TEMPO (5 mol%), and BiCl₃ (10 mol%) in 3 mL of **solvent** at 50 °C for 48 h with an O₂ balloon in a 25 mL flask. The NMR yield and recovery were determined by ¹H NMR analysis using dibromomethane as the internal standard. ^b The reaction was conducted at 30 °C.

Table S7. Optimization of the nitroxyl^a

$n\text{-C}_{15}\text{H}_{31}\text{CH}_2\text{OH}$ + MeOH 1a (1.0 mmol) 2e (5.0 equiv)		$\xrightarrow[\text{Fe(NO}_3)_3 \cdot 9\text{H}_2\text{O (6 mol\%)}]{\text{BiCl}_3 \text{ (10 mol\%)}nitroxyl \text{ (5 mol\%)}}$ Air + pure O_2 balloon DCE, 50 $^\circ\text{C}$, 48 h	$n\text{-C}_{15}\text{H}_{31}\text{CHO}$ + $\text{HCOO}n\text{-C}_{16}\text{H}_{33}$ + $n\text{-C}_{15}\text{H}_{31}\text{COOMe}$ 3a 11ea 6ae $n\text{-C}_{15}\text{H}_{31}\text{COO}n\text{-C}_{16}\text{H}_{33}$ +  5a 7ae				
Entry	nitroxyl	Recovery of 1a	NMR Y of 3a	NMR Y of 11ea	NMR Y of 6ae	NMR Y of 5a	NMR Y of 7ae
1	TEMPO	0	8%	1%	84%	4%	5%
2	4-NHAc-TEMPO	0	42%	0	21%	0	18%
3	4-OMe-TEMPO	0	11%	1%	81%	2%	3%
4	4-OH-TEMPO	0	33%	1%	40%	4%	18%
5	ABNO	3%	44%	2%	8%	0	22%

^a The reaction was conducted with **1a** (1.0 mmol), MeOH (5.0 mmol), Fe(NO₃)₃•9H₂O (6 mol%), **nitroxyl** (5 mol%), and BiCl₃ (10 mol%) in 3 mL of DCE at 50 °C for 48 h with an O₂ balloon in a 25 mL flask. The NMR yield and recovery were determined by ¹H NMR analysis using dibromomethane as the internal standard.

Table S8. Optimization of the Lewis acid^a

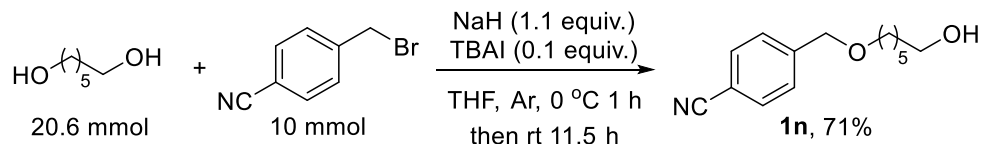
$n\text{-C}_{15}\text{H}_{31}\text{CH}_2\text{OH}$ + MeOH 1a (1.0 mmol) 2e (5.0 equiv)			$\xrightarrow[\text{Fe(NO}_3)_3 \cdot 9\text{H}_2\text{O (6 mol\%)}]{\text{LA (10 mol\%)}TEMPO (5 mol\%)}$ Air + pure O ₂ balloon DCE, 50 °C, 48 h		$n\text{-C}_{15}\text{H}_{31}\text{CHO}$ + $\text{HCOO}n\text{-C}_{16}\text{H}_{33}$ + $n\text{-C}_{15}\text{H}_{31}\text{COOMe}$ 3a 11ea 6ae $n\text{-C}_{15}\text{H}_{31}\text{COO}n\text{-C}_{16}\text{H}_{33}$ +  5a 7ae		
Entry	LA	Recovery of 1a	NMR Y of 3a	NMR Y of 11ea	NMR Y of 6ae	NMR Y of 5a	NMR Y of 7ae
1^b	BiCl₃	0	9%	1%	83% (73%)	2%	3%
2	FeCl ₃	0	33%	0	31%	4%	22%
3	InCl ₃	0	29%	15%	35%	8%	7%
4	InBr ₃	0	10%	6%	57%	8%	11%
5	BiBr ₃	26%	16%	1%	2%	2%	16%

^a The reaction was conducted with **1a** (1.0 mmol), MeOH (5.0 mmol), Fe(NO₃)₃•9H₂O (6 mol%), TEMPO (5 mol%), and **Lewis acid** (10 mol%) in 3 mL of DCE at 50 °C for 48 h with an O₂ balloon in a 50 mL flask. The NMR yield and recovery were determined by ¹H NMR analysis using dibromomethane as the internal standard. ^b 5% of *n*-C₁₅H₃₁COOH **4a** was isolated.

Experimental details and analytical data

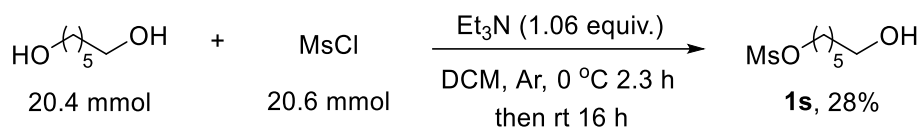
1. Preparation of alcohols

(1) Preparation of 4-(((6-hydroxyhexyl)oxy)methyl)benzonitrile (**1n**, yyb-7-005)



To a flask was added 1,6-hexanediol (2.4879 g, 20.6 mmol, 98% purity). Then the flask was degassed and refilled with argon for three times, the anhydrous THF (40 mL) was added and cooled to 0 °C with an ice-water bath before the addition of NaH (445.7 mg, 11 mmol, 60% in oil). After stirring for 1 h at the same temperature, 4-cyanobenzyl bromide (2.0057 g, 10 mmol, 98% purity) and TBAI (Tetrabutylammonium iodide, 373.9 mg, 1.0 mmol) were added. The resulting mixture was warmed to rt, stirred for 11.5 h, quenched with a saturated aqueous solution of NH₄Cl, and extracted with ethyl acetate (30 mL x 3). The combined organic layer was washed with brine and dried over anhydrous Na₂SO₄. After filtration and evaporation, the crude product was purified by column chromatography on silica gel to afford **1n** (1.6552 g, 71%) [eluent: petroleum ether/ethyl acetate = 10/1 (~330 mL) to 3/1 (~400 mL), then 2/1 (~600 mL)] as yellow oil. **¹H NMR** (400 MHz, CDCl₃) δ = 7.63 (d, *J* = 8.0 Hz, 2 H, Ar-H), 7.45 (d, *J* = 8.4 Hz, 2 H, Ar-H), 4.55 (s, 2 H, OCH₂), 3.62 (t, *J* = 6.4 Hz, 2 H, OCH₂), 3.50 (t, *J* = 6.6 Hz, 2 H, OCH₂), 2.18 (br, 1 H, OH), 1.65 (quint, *J* = 6.8 Hz, 2 H, CH₂), 1.57 (quint, *J* = 6.8 Hz, 2 H, CH₂), 1.47-1.32 (m, 4 H, 2 x CH₂); **¹³C NMR** (100 MHz, CDCl₃): δ = 144.2, 132.0, 127.5, 118.7, 110.9, 71.6, 70.8, 62.4, 32.4, 29.4, 25.8, 25.4; **IR** (neat, cm⁻¹): ν = 2934, 2858, 2228, 1610, 1408, 1361, 1172, 1096; **MS** (70 eV, EI) *m/z* (%): 233 (M⁺, 13.1), 116 (100); **HRMS** (EI) calcd *m/z* for C₁₄H₁₉NO₂ [M]⁺: 233.1410, Found: 233.1410.

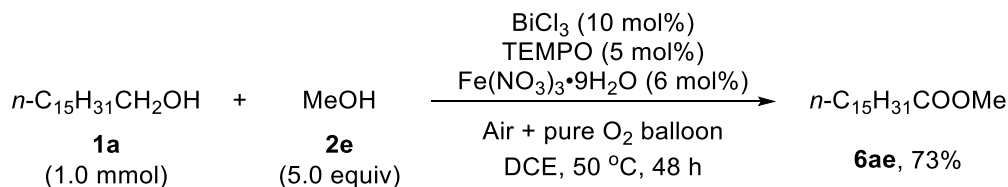
(2) Preparation of 6-hydroxyhexyl methanesulfonate¹ (**1s**, yyb-6-079)



To a flask were added 1,6-hexanediol (2.4841 g, 20.4 mmol, 97% purity), Et₃N (3.0 mL, d = 0.728 g/mL, 21.6 mmol), and DCM (40 mL). Then MsCl (1.6 mL, d = 1.478 g/mL, 20.6 mmol) was added dropwise within 20 min at 0 °C with an ice-water bath. After being stirred for 2 h at the same temperature, the resulting mixture was warmed to rt and stirred for 16 h. The solvent was removed under reduced pressure and the residue was purified by chromatography on silica gel to afford **1s** (1.1072 g, 28%) [first round eluent: petroleum ether/ethyl acetate = 5/1 (~300 mL) to 3/1 (~400 mL), then 2/1 (~600 mL) to 1/1 (~600 mL) to afford pure **1s** (141.5 mg) and impure **1s**; second round eluent: DCM/methol = 200/1 (~600 mL) to 100/1 (~400 mL), then 20/1 (~420 mL) to afford pure **1s** (0.9657 g)] as light yellow oil. ¹H NMR (400 MHz, CDCl₃) δ = 4.23 (t, *J* = 6.6 Hz, 2 H, OCH₂), 3.62 (t, *J* = 6.4 Hz, 2 H, OCH₂), 3.01 (s, 3 H, CH₃), 2.22 (s, 1 H, OH), 1.77 (quint, *J* = 6.6 Hz, 2 H, CH₂), 1.57 (quint, *J* = 6.6 Hz, 2 H, CH₂), 1.50-1.34 (m, 4 H, 2 x CH₂); ¹³C NMR (100 MHz, CDCl₃): δ = 70.0, 62.2, 37.1, 32.2, 28.9, 24.99, 24.97; IR (neat, cm⁻¹): ν = 2934, 2864, 1468, 1340, 1262, 1169, 1052, 1022; MS (ESI) *m/z*: 219 (M+Na)⁺, 197 (M+H)⁺; HRMS (ESI) calcd *m/z* for C₇H₁₇O₄S [M+H]⁺: 197.0842, found 197.0843.

2. Preparation of carboxylic acid esters

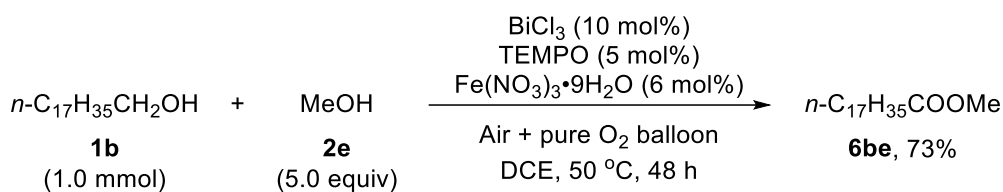
(1) Preparation of methyl palmitate (**6ae**, yyb-8-102)



Typical Procedure I: To a 50 mL flask were added Fe(NO₃)₃•9H₂O (24.9 mg, 0.06 mmol), TEMPO (8.0 mg, 0.05 mmol), BiCl₃ (31.4 mg, 0.1 mmol), **1a** (243.1 mg, 1.0 mmol), MeOH (202 μL, 5 mmol), and DCE (3 mL) sequentially in open air. The reaction was then carried out at 50 °C under the atmosphere of O₂ from a balloon (about

2.5 L) until completion of the reaction as monitored by TLC (48 h). The resulting mixture was filtrated through a short column of silica gel eluted with diethyl ether (3 x 25 mL) and concentrated. **6ae** was formed in 83% NMR yield based on ^1H NMR analysis of the crude product with CH_2Br_2 (35 μL) as the internal standard. After evaporation, MeOH (2 mL) and a saturated NaHSO_3 aqueous solution (1 mL) were added to the residue. The resulting mixture was stirred vigorously at room temperature for 1 h, dried with anhydrous Na_2SO_4 , filtrated through a short column of silica gel (1 cm) eluted with DCM (3 x 15 mL), and concentrated. After evaporation, the residue was purified by chromatography on silica gel to afford **6ae**² (198.3 mg, 73%) [eluent: petroleum ether/DCM = 10/1 (~220 mL) to 4/1 (~250 mL), 2/1 (~150 mL), then petroleum ether/ethyl acetate = 5/1 (~300 mL)] as light yellow solid (Low melting point solid, unable to recrystallize to measure melting point). ^1H NMR (400 MHz, CDCl_3): δ = 3.66 (s, 3 H, OCH_3), 2.30 (t, J = 7.4 Hz, 2 H, CH_2), 1.67-1.55 (m, 2 H, CH_2), 1.38-1.15 (m, 24 H, 12 x CH_2), 0.88 (t, J = 6.8 Hz, 3 H, CH_3); ^{13}C NMR (100 MHz, CDCl_3): δ = 174.2, 51.3, 34.0, 31.9, 29.65, 29.63, 29.62, 29.55, 29.4, 29.3, 29.2, 29.1, 24.9, 22.6, 14.0; IR (neat, cm^{-1}): ν = 2950, 2916, 2849, 1739, 1465, 1436, 1377, 1195, 1164; MS (70 eV, EI) m/z (%): 270 (M^+ , 19.23), 74 (100).

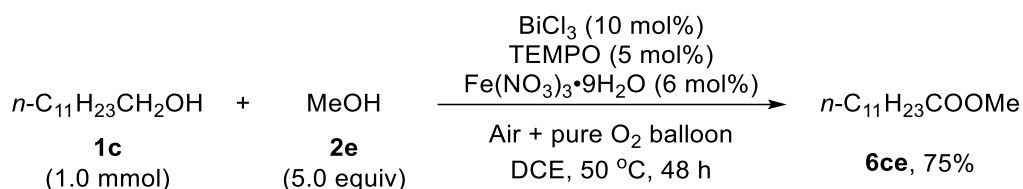
(2) Preparation of methyl stearate (**6be**, QAN-8-169)



Following Typical Procedure I, the reaction of **1b** (270.5mg, 1.0 mmol), $\text{Fe}(\text{NO}_3)_3 \cdot 9\text{H}_2\text{O}$ (24.7 mg, 0.06 mmol), TEMPO (8.2 mg, 0.05 mmol), BiCl_3 (31.6 mg, 0.1 mmol), and MeOH (202 μL , 5 mmol) in DCE (3 mL) afforded **6be**² (217.8 mg, 73%) [eluent: petroleum ether/DCM = 10/1 (~220 mL) to 4/1 (~250 mL), then 2/1 (~450 mL)] as white solid: m.p. 42.1-43.5 $^\circ\text{C}$ (We were not able to the recrystallize from the solvents tested, the m.p. of **6be** was determined by using the solid right after evaporation of the eluent). ^1H NMR (400 MHz, CDCl_3) δ = 3.66 (s, 3 H, OCH_3), 2.30 (t, J = 7.6 Hz, 2 H, CH_2), 1.65-1.58 (m, 2 H, CH_2), 1.32-1.17 (m, 28 H, 14 x CH_2), 0.88

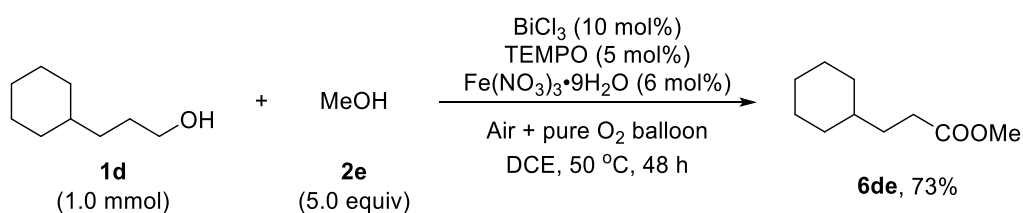
(t, $J = 6.4$ Hz, 3 H, CH₃); ¹³C NMR (100 MHz, CDCl₃) $\delta = 174.3, 51.3, 34.1, 31.9, 29.7, 29.6, 29.4, 29.3, 29.2, 29.1, 24.9, 22.7, 14.1$; IR (neat, cm⁻¹): $\nu = 2916, 2848, 1738, 1463, 1435, 1380, 1331, 1254, 1213, 1194, 1169, 1105$; MS (70 eV, EI) m/z (%): 298 (M⁺, 43.97), 74 (100).

(3) Preparation of methyl dodecanoate (**6ce**, lj-8-081)



Following Typical Procedure I, the reaction of **1c** (186.4 mg, 1.0 mmol), Fe(NO₃)₃•9H₂O (24.6 mg, 0.06 mmol), TEMPO (8.1 mg, 0.05 mmol), BiCl₃ (31.6 mg, 0.1 mmol), and MeOH (202 μ L, 5 mmol) in DCE (3 mL) afforded **6ce**² (160.7 mg, 75%) [eluent: petroleum ether/ethyl acetate = 20/1 (~420 mL)] as oil. ¹H NMR (400 MHz, CDCl₃) $\delta = 3.66$ (s, 3 H, OCH₃), 2.30 (t, $J = 7.6$ Hz, 2 H, CH₂), 1.70-1.57 (m, 2 H, CH₂), 1.45-1.15 (m, 16 H, 8 x CH₂), 0.88 (t, $J = 6.6$ Hz, 3 H, CH₃); ¹³C NMR (100 MHz, CDCl₃) $\delta = 174.2, 51.3, 34.1, 31.9, 29.7, 29.6, 29.4, 29.3, 29.2, 29.1, 24.9, 22.6, 14.0$; IR (neat, cm⁻¹): $\nu = 2924, 2855, 1742, 1463, 1438, 1363, 1236, 1198, 1171, 1116$; MS (70 eV, EI) m/z (%): 214 (M⁺, 4.71), 74 (100).

(4) Preparation of methyl 3-cyclohexylpropanoate (**6de**, yyb-7-092)



Typical Procedure II: To a 50 mL flask were added Fe(NO₃)₃•9H₂O (24.8 mg, 0.06 mmol), TEMPO (8.4 mg, 0.05 mmol), BiCl₃ (31.4 mg, 0.1 mmol), **1d** (141.7 mg, 1.0 mmol), MeOH (202 μ L, 5 mmol), and DCE (3 mL) sequentially in open air. The reaction was then carried out at 50 °C under the atmosphere of O₂ from a balloon (about 2.5 L) until completion of the reaction as monitored by TLC (48 h). The resulting mixture was filtrated through a short column of silica gel eluted with diethyl ether (3 x 25 mL) and concentrated. **6de** was formed in 83% NMR yield based on ¹H NMR

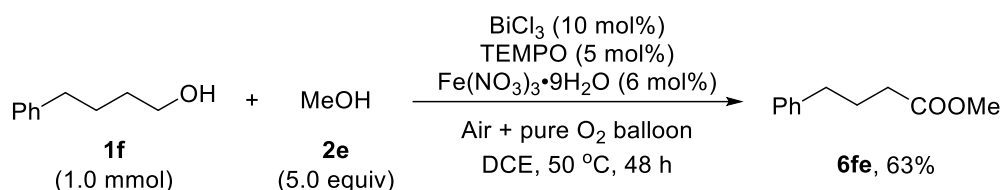
analysis of the crude product with CH₂Br₂ (35 μ L) as the internal standard. After evaporation, THF (2 mL) and 3 M HCl (0.5 mL) were added to the residue with vigorous stirring at room temperature for 2 h. To the resulting mixture was added a saturated aqueous solution of NaHCO₃ slowly until pH ~8. Then a saturated NaHSO₃ aqueous solution (2 mL) was added to the residue with vigorous stirring at room temperature for 2 h. The resulting mixture was extracted with DCM (15 mL $\times$ 3), washed with brine, and dried over anhydrous Na₂SO₄. After filtration and evaporation, the residue was purified by chromatography on silica gel to afford **6de**³ (122.0 mg, 73%) [eluent: petroleum ether/ethyl acetate = 100/1 (~300 mL) to 50/1 (~100 mL)] as yellow oil. ¹H NMR (400 MHz, CDCl₃): δ = 3.66 (s, 3 H, OCH₃), 2.32 (t, J = 7.8 Hz, 2 H, CH₂), 1.77-1.59 (m, 5 H), 1.52 (dd, J_1 = 15.4 Hz, J_2 = 7.0 Hz, 2 H, CH₂), 1.30-1.10 (m, 4 H), 0.96-0.82 (m, 2 H); ¹³C NMR (100 MHz, CDCl₃): δ = 174.5, 51.3, 37.1, 32.9, 32.3, 31.6, 26.4, 26.1; IR (neat, cm⁻¹): ν = 2922, 2851, 1739, 1449, 1436, 1366, 1195, 1165; MS (70 eV, EI) m/z (%): 170 (M⁺, 1.01), 97 (100).

(5) Preparation of methyl 3-phenylpropanoate (**6ee**, lj-8-082)



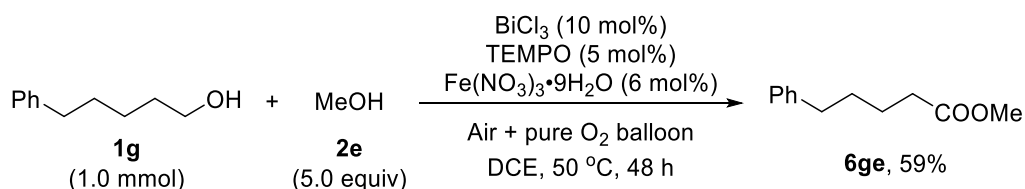
Following Typical Procedure I, the reaction of **1e** (136.3 mg, 1.0 mmol), Fe(NO₃)₃·9H₂O (24.6 mg, 0.06 mmol), TEMPO (8.1 mg, 0.05 mmol), BiCl₃ (31.3 mg, 0.1 mmol), and MeOH (202 μ L, 5 mmol) in DCE (3 mL) afforded **6ee**⁴ (105.1 mg, 64%) [eluent: petroleum ether/ethyl acetate = 20/1 (~420 mL)] as oil. ¹H NMR (400 MHz, CDCl₃): δ = 7.32-7.25 (m, 2 H, Ar-H), 7.24-7.12 (m, 3 H, Ar-H), 3.67 (s, 3 H, OCH₃), 2.95 (t, J = 7.8 Hz, 2 H, CH₂), 2.64 (t, J = 7.8 Hz, 2 H, CH₂); ¹³C NMR (100 MHz, CDCl₃): δ = 173.3, 140.5, 128.5, 128.2, 126.2, 51.5, 35.7, 30.9; IR (neat, cm⁻¹): ν = 3026, 2950, 2834, 1736, 1496, 1443, 1364, 1290, 1255, 1164, 1075, 1055, 1028; MS (70 eV, EI) m/z (%): 164 (M⁺, 38.9), 104 (100).

(6) Preparation of methyl 4-phenylbutanoate (**6fe**, yyb-6-060)



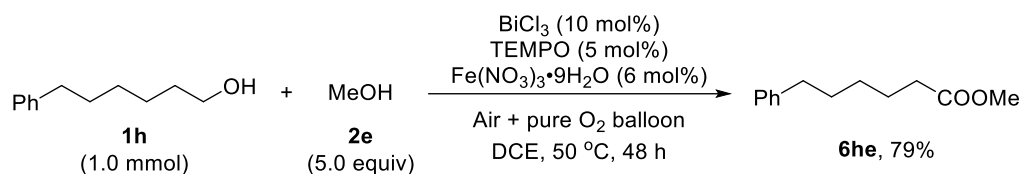
Following Typical Procedure I, the reaction of **1f** (149.5 mg, 1.0 mmol), Fe(NO₃)₃•9H₂O (24.9 mg, 0.06 mmol), TEMPO (8.2 mg, 0.05 mmol), BiCl₃ (31.4 mg, 0.1 mmol), and MeOH (202 μL, 5 mmol) in DCE (3 mL) afforded **6fe**³ (115.6 mg, 63%, 96% purity) [eluent: petroleum ether/DCM = 10/1 (~220 mL) to 4/1 (~250 mL), then 2/1 (~300 mL)] as light yellow oil. ¹H NMR (400 MHz, CDCl₃): δ = 7.26 (t, *J* = 7.4 Hz, 2 H, Ar-H), 7.21-7.10 (m, 3 H, Ar-H), 3.64 (s, 3 H, OCH₃), 2.64 (t, *J* = 7.6 Hz, 2 H, CH₂), 2.31 (t, *J* = 7.6 Hz, 2 H, CH₂), 1.95 (quint, *J* = 7.5 Hz, 2 H, CH₂); ¹³C NMR (100 MHz, CDCl₃): δ = 173.8, 141.3, 128.4, 128.3, 125.9, 51.3, 35.0, 33.3, 26.4; IR (neat, cm⁻¹): ν = 3027, 2946, 2864, 1735, 1496, 1442, 1366, 1246, 1168, 1146; MS (70 eV, EI) *m/z* (%): 178 (M⁺, 42.91), 104 (100).

(7) Preparation of methyl 5-phenylpentanoate (**6ge**, yyb-6-061)



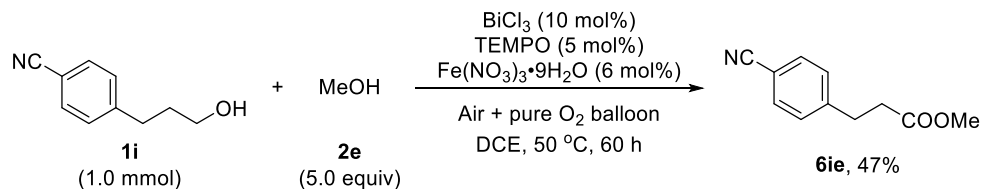
Following Typical Procedure I, the reaction of **1g** (167.5 mg, 1.0 mmol, 98% purity), Fe(NO₃)₃•9H₂O (24.7 mg, 0.06 mmol), TEMPO (8.1 mg, 0.05 mmol), BiCl₃ (32.1 mg, 0.1 mmol), and MeOH (202 μL, 5 mmol) in DCE (3 mL) afforded **6ge**³ (111.8 mg, 59%) [eluent: petroleum ether/DCM = 10/1 (~220 mL) to 4/1 (~250 mL), then 2/1 (~300 mL)] as light yellow oil. ¹H NMR (400 MHz, CDCl₃): δ = 7.26 (t, *J* = 7.4 Hz, 2 H, Ar-H), 7.20-7.12 (m, 3 H, Ar-H), 3.65 (s, 3 H, OCH₃), 2.62 (t, *J* = 6.6 Hz, 2 H, CH₂), 2.32 (t, *J* = 6.6 Hz, 2 H, CH₂), 1.74-1.55 (m, 4 H, 2 x CH₂); ¹³C NMR (100 MHz, CDCl₃): δ = 174.0, 142.1, 128.32, 128.26, 125.7, 51.4, 35.5, 33.9, 30.8, 24.5; IR (neat, cm⁻¹): ν = 3026, 2943, 2861, 1736, 1495, 1436, 1360, 1198, 1142; MS (70 eV, EI) *m/z* (%): 192 (M⁺, 3.56), 91 (100).

(8) Preparation of methyl 6-phenylhexanoate (**6he**, yyb-6-096)



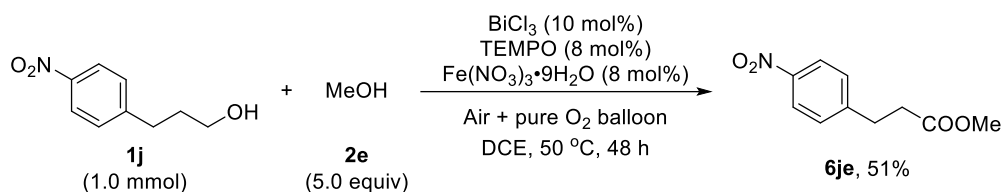
Following Typical Procedure I, the reaction of **1h** (182.8 mg, 97% purity, 1.0 mmol), $\text{Fe}(\text{NO}_3)_3 \cdot 9\text{H}_2\text{O}$ (24.5 mg, 0.06 mmol), TEMPO (8.2 mg, 0.05 mmol), BiCl_3 (31.3 mg, 0.1 mmol), and MeOH (202 μL , 5 mmol) in DCE (3 mL) afforded **6he**⁵ (162.6 mg, 79%) [eluent: petroleum ether/DCM = 15/1 (~320 mL) to 5/1 (~480 mL), then 3/1 (~400 mL)] as light yellow oil. **¹H NMR** (400 MHz, CDCl_3): δ = 7.25 (t, J = 7.2 Hz, 2 H, Ar-H), 7.20-7.09 (m, 3 H, Ar-H), 3.64 (s, 3 H, OCH_3), 2.59 (t, J = 7.6 Hz, 2 H, CH_2), 2.28 (t, J = 7.6 Hz, 2 H, CH_2), 1.71-1.56 (m, 4 H, 2 x CH_2), 1.35 (quint, J = 7.6 Hz, 2 H, CH_2); **¹³C NMR** (100 MHz, CDCl_3): δ = 174.0, 142.4, 128.3, 128.2, 125.6, 51.3, 35.6, 33.9, 30.9, 28.6, 24.7; **IR** (neat, cm^{-1}): ν = 3026, 2930, 2857, 1736, 1495, 1436, 1362, 1198, 1170; **MS** (70 eV, EI) m/z (%): 206 (M^+ , 9.32), 91 (100).

(9) Preparation of methyl 3-(4-cyanophenyl)propanoate (**6ie**, yyb-7-024)



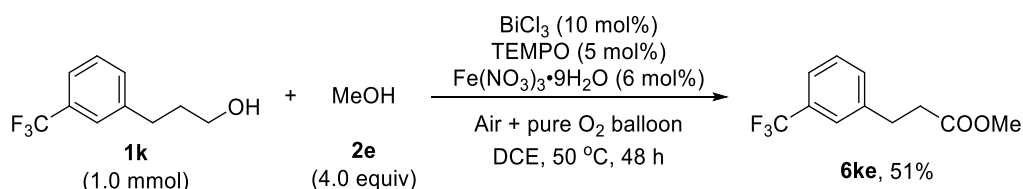
Following Typical Procedure II, the reaction of **1i** (161.1 mg, 1.0 mmol), $\text{Fe}(\text{NO}_3)_3 \cdot 9\text{H}_2\text{O}$ (24.6 mg, 0.06 mmol), TEMPO (8.2 mg, 0.05 mmol), BiCl_3 (31.3 mg, 0.1 mmol), and MeOH (202 μL , 5 mmol) in DCE (4 mL) afforded **6ie**⁶ (89.7 mg, 47%) [eluent: petroleum ether/ethyl acetate = 10/1 (~270 mL) to 8/1 (~450 mL)] as yellow oil. **¹H NMR** (400 MHz, CDCl_3): δ = 7.58 (d, J = 8.4 Hz, 2 H, Ar-H), 7.32 (d, J = 8.4 Hz, 2 H, Ar-H), 3.67 (s, 3 H, OCH_3), 3.02 (t, J = 7.6 Hz, 2 H, CH_2), 2.66 (t, J = 7.6 Hz, 2 H, CH_2); **¹³C NMR** (100 MHz, CDCl_3): δ = 172.5, 146.0, 132.2, 129.1, 118.8, 110.1, 51.6, 34.7, 30.7; **IR** (neat, cm^{-1}): ν = 2952, 2227, 1733, 1608, 1506, 1437, 1365, 1174, 1159; **MS** (70 eV, EI) m/z (%): 189 (M^+ , 21.63), 129 (100).

(10) Preparation of methyl 3-(4-nitrophenyl)propanoate (**6je**, yyb-7-049)



Following Typical Procedure II, the reaction of **1j** (181.3 mg, 1.0 mmol), $\text{Fe}(\text{NO}_3)_3 \cdot 9\text{H}_2\text{O}$ (32.5 mg, 0.08 mmol), TEMPO (13.0 mg, 0.08 mmol), BiCl_3 (31.3 mg, 0.1 mmol), and MeOH (202 μL , 5 mmol) in DCE (4 mL) afforded **6je**⁷ (106.4 mg, 51%) [eluent: petroleum ether/ethyl acetate = 7/1 (~400 mL) to 5/1 (~120 mL)] as light yellow solid; m.p. 73.7-74.2 °C (petroleum ether/DCM). ¹H NMR (400 MHz, CDCl_3): δ = 8.15 (d, J = 8.4 Hz, 2 H, Ar-H), 7.38 (d, J = 8.4 Hz, 2 H, Ar-H), 3.68 (s, 3 H, OCH_3), 3.07 (t, J = 7.4 Hz, 2 H, CH_2), 2.69 (t, J = 7.4 Hz, 2 H, CH_2); ¹³C NMR (100 MHz, CDCl_3): δ = 172.5, 148.2, 146.6, 129.1, 123.7, 51.7, 34.7, 30.5; IR (neat, cm^{-1}): ν = 3113, 3083, 1727, 1607, 1514, 1430, 1346, 1191, 1169; MS (70 eV, EI) m/z (%): 209 (M^+ , 26.18), 149 (100).

(11) Preparation of methyl 3-(3-(trifluoromethyl)phenyl)propanoate (**6ke**, yyb-7-072)



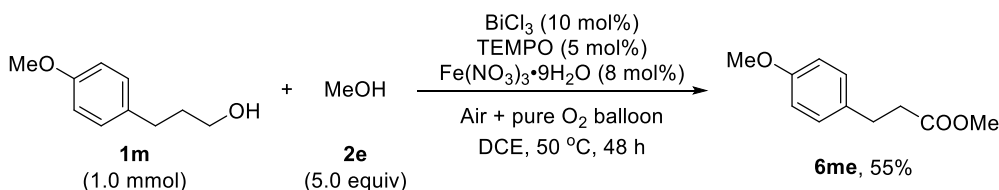
Following Typical Procedure II, the reaction of **1k** (204.1 mg, 1.0 mmol), $\text{Fe}(\text{NO}_3)_3 \cdot 9\text{H}_2\text{O}$ (24.7 mg, 0.06 mmol), TEMPO (8.3 mg, 0.05 mmol), BiCl_3 (31.4 mg, 0.1 mmol), and MeOH (162 μL , 4 mmol) in DCE (4 mL) afforded **6ke**⁶ (121.8 mg, 51%, 98% purity) [eluent: petroleum ether/ethyl acetate = 20/1 (~420 mL)] as light yellow oil. ¹H NMR (400 MHz, CDCl_3): δ = 7.51-7.44 (m, 2 H, Ar-H), 7.42-7.34 (m, 2 H, Ar-H), 3.67 (s, 3 H, OCH_3), 3.01 (t, J = 7.8 Hz, 2 H, CH_2), 2.66 (t, J = 7.6 Hz, 2 H, CH_2); ¹³C NMR (100 MHz, CDCl_3): δ = 172.8, 141.4, 131.7, 130.8 (q, J = 31.9 Hz), 128.9, 125.0 (q, J = 3.7 Hz), 124.1 (q, J = 270.4 Hz), 123.2 (q, J = 3.7 Hz), 51.6, 35.3, 30.6; ¹⁹F NMR (376 MHz, CDCl_3): δ = -63.1; IR (neat, cm^{-1}): ν = 1737, 1439, 1365, 1327, 1199, 1159, 1118, 1073; MS (70 eV, EI) m/z (%): 232 (M^+ , 44.2), 172 (100).

(12) Preparation of methyl 3-(4-chlorophenyl)propanoate (**6le**, yyb-7-091)



Following Typical Procedure II, the reaction of **1l** (169.9 mg, 1.0 mmol), $\text{Fe}(\text{NO}_3)_3 \cdot 9\text{H}_2\text{O}$ (24.7 mg, 0.06 mmol), TEMPO (8.1 mg, 0.05 mmol), BiCl_3 (31.4 mg, 0.1 mmol), and MeOH (202 μL , 5 mmol) in DCE (4 mL) afforded **6le**⁴ (103.2 mg, 52%) [eluent: petroleum ether/ethyl acetate = 25/1 (~260 mL) to 20/1 (~100 mL)] as yellow oil. **¹H NMR** (400 MHz, CDCl_3): δ = 7.30-7.20 (m, 2 H, Ar-H), 7.17-7.07 (m, 2 H, Ar-H), 3.66 (s, 3 H, OCH_3), 2.91 (t, J = 7.6 Hz, 2 H, CH_2), 2.60 (t, J = 7.8 Hz, 2 H, CH_2); **¹³C NMR** (100 MHz, CDCl_3): δ = 173.0, 138.9, 132.0, 129.6, 128.5, 51.6, 35.4, 30.2; **IR** (neat, cm^{-1}): ν = 2951, 1734, 1492, 1436, 1364, 1195, 1157, 1092; **MS** (70 eV, EI) m/z (%): 200 ($\text{M}^{(37}\text{Cl})^+$, 9.7), 198 ($\text{M}^{(35}\text{Cl})^+$, 29.17), 138 (100).

(13) Preparation of methyl 3-(4-methoxyphenyl)propanoate (**6me**, yyb-7-063)



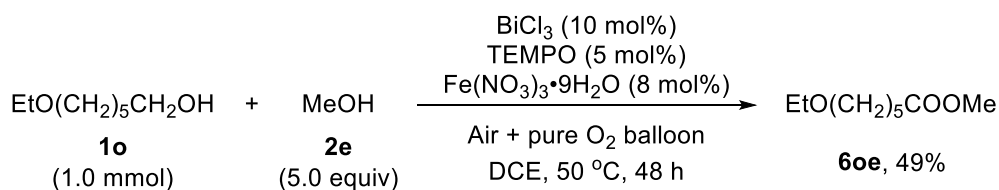
Following Typical Procedure II, the reaction of **1m** (175.5 mg, 95% purity, 1.0 mmol), $\text{Fe}(\text{NO}_3)_3 \cdot 9\text{H}_2\text{O}$ (32.3 mg, 0.08 mmol), TEMPO (8.1 mg, 0.05 mmol), BiCl_3 (32.0 mg, 0.1 mmol), and MeOH (202 μL , 5 mmol) in DCE (4 mL) afforded **6me**⁶ (107.3 mg, 55%) [eluent: petroleum ether/ethyl acetate = 15/1 (~320 mL) to 10/1 (~110 mL)] as light yellow oil. **¹H NMR** (400 MHz, CDCl_3): δ = 7.10 (d, J = 8.8 Hz, 2 H, Ar-H), 6.82 (d, J = 8.4 Hz, 2 H, Ar-H), 3.76 (s, 3 H, OCH_3), 3.65 (s, 3 H, OCH_3), 2.88 (t, J = 7.8 Hz, 2 H, CH_2), 2.59 (t, J = 7.8 Hz, 2 H, CH_2); **¹³C NMR** (100 MHz, CDCl_3): δ = 173.3, 158.0, 132.5, 129.1, 113.8, 55.1, 51.4, 35.9, 30.0; **IR** (neat, cm^{-1}): ν = 3011, 2929, 1729, 1610, 1511, 1436, 1373, 1176, 1157; **MS** (70 eV, EI) m/z (%): 194 (M^+ , 21.47), 121 (100).

(14) Preparation of methyl 6-(4-cyanobenzyloxy)hexanoate (**6ne**, yyb-7-015)



Following Typical Procedure II, the reaction of **1n** (233.0 mg, 1.0 mmol), $\text{Fe}(\text{NO}_3)_3 \cdot 9\text{H}_2\text{O}$ (24.4 mg, 0.06 mmol), TEMPO (8.3 mg, 0.05 mmol), BiCl_3 (31.8 mg, 0.1 mmol), and MeOH (202 μL , 5 mmol) in DCE (3 mL) afforded **6ne** (129.9 mg, 50%) [eluent: petroleum ether/ ethyl acetate = 10/1 (~330 mL) to 5/1 (~240 mL)] as light yellow oil. ^1H NMR (400 MHz, CDCl_3): δ = 7.63 (d, J = 8.0 Hz, 2 H, Ar-H), 7.44 (d, J = 8.0 Hz, 2 H, Ar-H), 4.54 (s, 2 H, OCH_2), 3.67 (s, 3 H, OCH_3), 3.50 (t, J = 6.6 Hz, 2 H, OCH_2), 2.33 (t, J = 7.6 Hz, 2 H, CH_2), 1.71-1.59 (m, 4 H, 2 x CH_2), 1.48-1.35 (m, 2 H, CH_2); ^{13}C NMR (100 MHz, CDCl_3): δ = 173.9, 144.2, 132.0, 127.5, 118.7, 111.0, 71.7, 70.6, 51.3, 33.8, 29.2, 25.6, 24.5; IR (neat, cm^{-1}): ν = 2947, 2863, 2223, 1738, 1609, 1448, 1371, 1236, 1170, 1099; MS (70 eV, EI) m/z (%): 261 (M^+ , 3.52), 116 (100); HRMS (EI) calcd m/z for $\text{C}_{15}\text{H}_{19}\text{NO}_3$ [M] $^+$: 261.1359, Found: 261.1363.

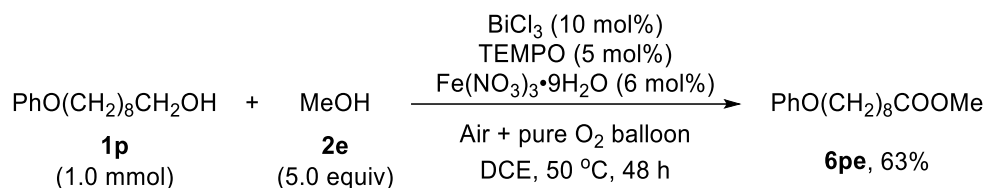
(15) Preparation of methyl 6-ethoxyhexanoate (**6oe**, yyb-6-155)



Following Typical Procedure II, the reaction of **1o** (146.0 mg, 1.0 mmol), $\text{Fe}(\text{NO}_3)_3 \cdot 9\text{H}_2\text{O}$ (32.6 mg, 0.08 mmol), TEMPO (8.1 mg, 0.05 mmol), BiCl_3 (31.8 mg, 0.1 mmol), and MeOH (202 μL , 5 mmol) in DCE (3 mL) afforded **6oe** (88.6 mg, 49%, 96% purity) [eluent: petroleum ether/ethyl acetate = 20/1 (~210 mL) to 10/1 (~220 mL)] as light yellow oil. ^1H NMR (400 MHz, CDCl_3): δ = 3.66 (s, 3 H, OCH_3), 3.46 (q, J = 6.9 Hz, 2 H, OCH_2), 3.41 (t, J = 6.6 Hz, 2 H, OCH_2), 2.32 (t, J = 7.6 Hz, 2 H, CH_2), 1.73-1.52 (m, 4 H, 2 x CH_2), 1.46-1.32 (m, 2 H, CH_2), 1.19 (t, J = 7.0 Hz, 3 H, CH_3); ^{13}C NMR (100 MHz, CDCl_3): δ = 174.0, 70.2, 66.0, 51.3, 33.9, 29.3, 25.7, 24.7, 15.0; IR (neat, cm^{-1}): ν = 2975, 2940, 2860, 1738, 1437, 1376, 1201, 1167, 1109; MS (ESI) m/z : 197 ($\text{M}+\text{Na}$) $^+$, 175 ($\text{M}+\text{H}$) $^+$; HRMS (ESI) calcd m/z for $\text{C}_9\text{H}_{19}\text{O}_3$ [$\text{M}+\text{H}$] $^+$:

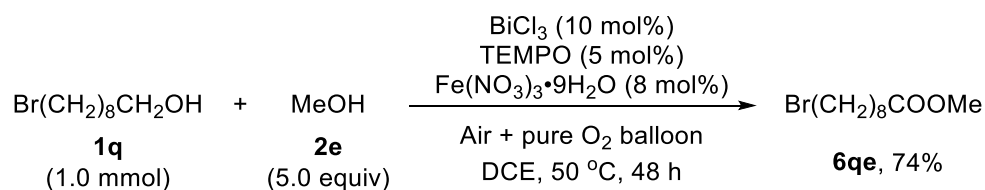
175.1329, found 175.1330.

(16) Preparation of methyl 9-phenoxy-nonanoate (6pe, yyb-6-172)



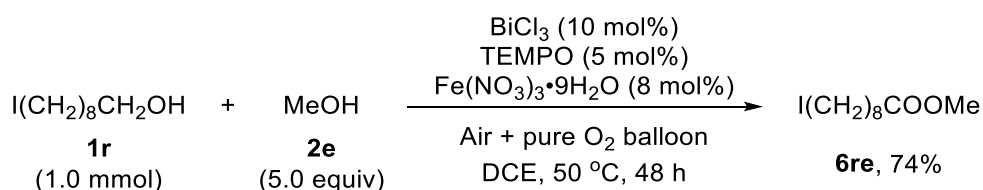
Typical Procedure III: To a 50 mL flask were added Fe(NO₃)₃•9H₂O (24.5 mg, 0.06 mmol), TEMPO (8.0 mg, 0.05 mmol), BiCl₃ (31.9 mg, 0.1 mmol), **1p** (236.7 mg, 1.0 mmol), MeOH (202 μL, 5 mmol), and DCE (3 mL) sequentially in open air. The reaction was then carried out at 50 °C under the atmosphere of O₂ from a balloon (about 2.5 L) until completion of the reaction as monitored by TLC (48 h). The resulting mixture was filtrated through a short column of silica gel eluted with diethyl ether (3 x 25 mL) and concentrated. **6pe** was formed in 73% NMR yield based on ¹H NMR analysis of the crude product with CH₂Br₂ (35 μL) as the internal standard. After evaporation, the residue was purified by chromatography on silica gel to afford **6pe** (166.4 mg, 63%) [first round eluent: petroleum ether/ethyl acetate = 40/1 (~610 mL); second round eluent: petroleum ether/ethyl acetate = 50/1 (~510 mL)] as light yellow oil. **¹H NMR** (400 MHz, CDCl₃): δ = 7.32-7.20 (m, 2 H, Ar-H), 6.96-6.83 (m, 3 H, Ar-H), 3.93 (t, *J* = 6.6 Hz, 2 H, OCH₂), 3.65 (s, 3 H, OCH₃), 2.29 (t, *J* = 7.6 Hz, 2 H, CH₂), 1.76 (quint, *J* = 7.0 Hz, 2 H, CH₂), 1.62 (quint, *J* = 7.3 Hz, 2 H, CH₂), 1.52-1.40 (m, 2 H, CH₂), 1.38-1.27 (m, 6 H, 3 x CH₂); **¹³C NMR** (100 MHz, CDCl₃): δ = 174.1, 159.0, 129.3, 120.3, 114.4, 67.7, 51.3, 33.9, 29.2, 29.1, 29.0, 25.9, 24.8; **IR** (neat, cm⁻¹): ν = 2930, 2856, 1737, 1599, 1496, 1470, 1436, 1242, 1198, 1169; **MS** (ESI) *m/z*: 287 (M+Na)⁺, 265 (M+H)⁺; **HRMS** (ESI) calcd *m/z* for C₁₆H₂₅O₃ [M+H]⁺: 265.1798, found 265.1796.

(17) Preparation of methyl 9-bromononanoate (6qe, yyb-6-085)



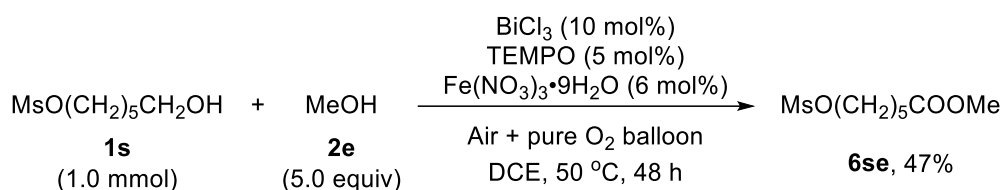
Following Typical Procedure I, the reaction of **1q** (223.5 mg, 1.0 mmol), Fe(NO₃)₃•9H₂O (32.3 mg, 0.08 mmol), TEMPO (8.3 mg, 0.05 mmol), BiCl₃ (31.8 mg, 0.1 mmol), and MeOH (202 μL, 5 mmol) in DCE (3 mL) afforded **6qe**⁸ (189.5 mg, 74%, 98% purity) [eluent: petroleum ether/DCM = 10/1 (~220 mL) to 4/1 (~250 mL), then 2/1 (~300 mL)] as light yellow oil. ¹H NMR (400 MHz, CDCl₃): δ = 3.66 (s, 3 H, OCH₃), 3.40 (t, *J* = 6.8 Hz, 2 H, CH₂), 2.30 (t, *J* = 7.4 Hz, 2 H, CH₂), 1.85 (quint, *J* = 7.1 Hz, 2 H, CH₂), 1.68-1.56 (m, 2 H, CH₂), 1.48-1.37 (m, 2 H, CH₂), 1.37-1.27 (m, 3 x CH₂); ¹³C NMR (100 MHz, CDCl₃): δ = 174.0, 51.3, 33.9, 33.7, 32.7, 28.91, 28.87, 28.4, 28.0, 24.7; IR (neat, cm⁻¹): ν = 2930, 2855, 1737, 1458, 1437, 1362, 1197, 1170; MS (ESI) *m/z*: 275 (M(⁸¹Br)+Na)⁺, 273 (M(⁷⁹Br)+Na)⁺.

(18) Preparation of methyl 9-iodononanoate (**6re**, yyb-6-088)



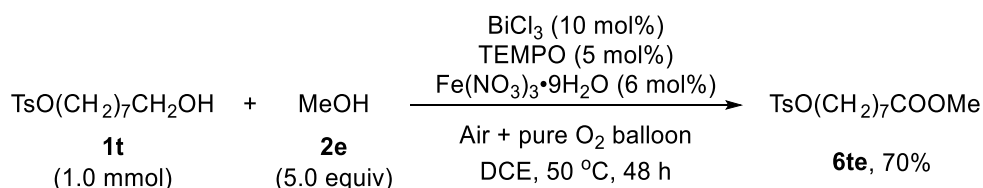
Following Typical Procedure I, the reaction of **1r** (269.9 mg, 1.0 mmol), Fe(NO₃)₃•9H₂O (32.3 mg, 0.08 mmol), TEMPO (8.0 mg, 0.05 mmol), BiCl₃ (32.1 mg, 0.1 mmol), and MeOH (202 μL, 5 mmol) in DCE (3 mL) afforded **6re**⁹ (200.1 mg and 23.2 mg (90% purity), 74%) [eluent: petroleum ether/DCM = 10/1 (~330 mL) to 4/1 (~250 mL), then 3/1 (~200 mL)] as light yellow oil. ¹H NMR (400 MHz, CDCl₃): δ = 3.67 (s, 3 H, OCH₃), 3.18 (t, *J* = 7.0 Hz, 2 H, CH₂), 2.30 (t, *J* = 7.6 Hz, 2 H, CH₂), 1.82 (quint, *J* = 7.1 Hz, 2 H, CH₂), 1.68-1.56 (m, 2 H, CH₂), 1.45-1.35 (m, 2 H, CH₂), 1.35-1.24 (m, 6 H, 3 x CH₂); ¹³C NMR (100 MHz, CDCl₃): δ = 174.1, 51.4, 34.0, 33.4, 30.3, 28.9, 28.2, 24.8, 7.1; IR (neat, cm⁻¹): ν = 2926, 2853, 1736, 1459, 1434, 1361, 1195, 1169; MS (ESI) *m/z*: 321 (M+Na)⁺, 299 (M+H)⁺.

(19) Preparation of methyl 6-((methanesulfonyl)oxy)hexanoate (6se, yyb-6-156)



Following Typical Procedure II, the reaction of **1s** (196.7 mg, 1.0 mmol), $\text{Fe(NO}_3)_3 \cdot 9\text{H}_2\text{O}$ (24.6 mg, 0.06 mmol), TEMPO (8.2 mg, 0.05 mmol), BiCl_3 (31.2 mg, 0.1 mmol), and MeOH (202 μL , 5 mmol) in DCE (3 mL) afforded **6se**¹⁰ (106.3 mg, 47%) [eluent: petroleum ether/ethyl acetate = 3/1 (~200 mL) to 2/1 (~300 mL)] as light yellow oil. ¹H NMR (400 MHz, CDCl_3): δ = 4.23 (t, J = 6.4 Hz, 2 H, OCH_2), 3.67 (s, 3 H, OCH_3), 3.01 (s, 3 H, CH_3), 2.34 (t, J = 7.4 Hz, 2 H, CH_2), 1.82-1.73 (m, 2 H, CH_2), 1.72-1.62 (m, 2 H, CH_2), 1.50-1.39 (m, 2 H, CH_2); ¹³C NMR (100 MHz, CDCl_3): δ = 173.7, 69.6, 51.4, 37.1, 33.5, 28.6, 24.8, 24.1; IR (neat, cm^{-1}): ν = 2935, 2859, 1732, 1438, 1349, 1169, 1104; MS (ESI) m/z : 247 ($\text{M}+\text{Na}$)⁺, 225 ($\text{M}+\text{H}$)⁺.

(20) Preparation of methyl 8-(tosyloxy)octanoate (6te, QAN-8-170)



Following Typical Procedure I, the reaction of **1t** (300.5mg, 1.0 mmol), $\text{Fe(NO}_3)_3 \cdot 9\text{H}_2\text{O}$ (24.5 mg, 0.06 mmol), TEMPO (8.1 mg, 0.05 mmol), BiCl_3 (31.8 mg, 0.1 mmol), and MeOH (202 μL , 5 mmol) in DCE (3 mL) afforded **6te** (230.1 mg, 70%) [eluent: petroleum ether/ ethyl acetate = 15/1 (~300 mL) to 10/1 (~400 mL)] as oil. ¹H NMR (400 MHz, CDCl_3): δ = 7.79 (d, J = 8.0 Hz, 2 H, Ar-H), 7.35 (d, J = 7.6 Hz, 2 H, Ar-H), 4.01 (t, J = 6.4 Hz, 2 H, CH_2), 3.66 (s, 3 H, OCH_3), 2.45 (s, 3 H, CH_3), 2.28 (d, J = 7.6 Hz, 2 H, CH_2), 1.68-1.50 (m, 4 H, 2 x CH_2), 1.34-1.18 (m, 6 H, 3 x CH_2); ¹³C NMR (100 MHz, CDCl_3): δ = 174.0, 144.6, 133.2, 129.7, 127.8, 70.5, 51.3, 33.8, 28.7, 28.6, 28.4, 25.0, 24.6, 21.5; IR (neat, cm^{-1}): ν = 2939, 2910, 2868, 1731, 1598, 1466, 1434, 1347, 1311, 1248, 1215, 1173, 1098, 1071, 1048; MS (70 eV, EI) m/z (%): 328 (M^+ , 7.42), 91 (100); HRMS (EI) calcd m/z for $\text{C}_{16}\text{H}_{24}\text{O}_5\text{S}$ [M]⁺: 328.1339, Found: 328.1339.